Logistics & Transportation

The full-domain reference · mrsupplychain.ai

The physical movement engine — from carrier selection to last-mile execution.

Logistics & Transportation is the discipline that decides how product physically reaches the customer, what that movement truly costs, and who is accountable when it fails. Every mode, carrier and route choice is a standing bet on cost against service. Inventory decides how much capital sits still; logistics decides what it costs to make it move — and whether the promise made to the customer survives contact with the road.

HOW TO READ THIS DOMAIN

1 · Freight Management & Mode Selection. Choosing how cargo moves, on what terms, and at what true cost

2 · Carrier Management & TMS. The systems, carriers and routes that keep or break the OTIF promise

3 · Cold Chain & Specialized Logistics. Segments where the standard rules do not apply

4 · Network Design & 3PL Management. Decisions that set structural cost and service for the next 3–5 years

5 · Logistics Technology & Innovation. From labor-intensive operation to data-driven system — without over-investing

STAGE 1 · COMMERCIAL · 01 OF 06

Freight Management & Mode Selection

The commercial layer of logistics — choosing how cargo moves, negotiating the terms under which it moves, and managing the cost to move it as a lever of competitive advantage rather than an invoice to be paid.

01 Transportation mode selection: road, rail, air & ocean

Mode selection is the decision of how each lane moves — road (FTL/LTL), rail or intermodal, air, or ocean — based on the product's value density, the service the customer was promised, and the total landed cost, not the freight rate alone.

Mode selection is a total-cost decision, not a freight-price decision

The correct mode minimizes total landed cost: freight + in-transit inventory cost (WACC × value × days) + additional safety stock required by lead-time variability + risk of stockout or obsolescence. For high-value, low-weight products, air is frequently cheaper on a TCO basis than ocean despite a 4–8× freight premium.

The four primary modes and their structural trade-offs

Road: maximum flexibility, ideal for <1,500 km and JIT. Dominant mode in Mexico for domestic and nearshore flows.**Rail**: lowest cost per ton-km for bulk and long-haul, limited flexibility and last-mile dependency.**Ocean**: most economical per unit for long-haul, but 20–60 days of transit capital and high variability.**Air**: highest cost, justified by value density, urgency, short shelf life, or product-launch windows.

Nearshoring changes the mode equation permanently

Moving production from Asia to Mexico converts the dominant long-haul mode from ocean (30–45 days) to road (1–4 days). The reduction in in-transit inventory and safety stock frequently exceeds the manufacturing cost differential.

Intermediate vs Advanced

Intermediate: applies standard mode criteria to assigned routes.**Advanced**: designs the mode strategy by product family and corridor using full TCO; runs annual mode strategy reviews when freight rates or product value change materially.

BEST PRACTICES

Always run the full TCO, never freight price alone

The visible freight line is frequently the smaller part of the true cost for high-value products.

Re-evaluate mode strategy after every major nearshoring or product-value shift

Mode decisions made under Asia-centric lead times become wrong the moment production moves to Mexico or product ASP rises significantly.

Use air strategically for NPI and high-obsolescence windows

Air at launch -> ocean as the product matures is a high-ROI dynamic mode strategy.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — High-value electronics components from Asia — Product value $420/kg, 45-day ocean transit, 10% WACC.

TCO Component	Ocean	Air
Freight	$0.95/kg	$5.13/kg
In-transit inventory cost	$5.18/kg	$0.35/kg
Total TCO	$6.13/kg	$5.48/kg

Result: Air is $0.65/kg cheaper on TCO. At 2.4 M kg/year the annual advantage is $1.56 M. For standard components at $18/kg the ranking reverses completely.

WHAT YOU WOULD USE

TMS & Mode Selection

SAP TM — Native mode and carrier optimization engine inside S/4HANA.

Blue Yonder TMS / Oracle OTM — Strong multi-modal rate management and optimization.

Rate Intelligence

Xeneta / Freightos — Ocean and air rate benchmarking platforms.

HOW IT IS MEASURED

Transportation Cost per Unit (full TCO)

Freight + in-transit inventory cost + insurance / Units moved.

Benchmark: Track trend by mode and corridor; absolute number is less useful than directional change after mode shifts.

Modal Mix %

% of volume (tons or value) by mode.

Benchmark: Optimal mix is the output of a TCO model, not an industry average.

> **Choose mode on total landed cost, not on rate.** Segment lanes, price the inventory each mode creates, and treat recurring premium freight as a planning failure to fix — not a transport cost to accept.

02 Incoterms, freight contracts & rate management

The commercial frame of transport: Incoterms 2020 define where risk and cost transfer between buyer and seller, freight contracts set rates and service commitments with carriers, and rate management keeps those prices current through tenders, indexation and audit.

Incoterms define who controls cost, risk and leverage

Incoterms allocate responsibility for freight, insurance, customs and risk between buyer and seller. The wrong Incoterm transfers control of freight to the party that has the least incentive to optimize it. Under CIF the supplier books the freight and optimizes for their own convenience and margin. Under FOB or EXW the buyer controls the freight and can consolidate volume across multiple suppliers.

The CIF -> FOB transition is one of the highest-ROI logistics moves

A large buyer consolidating Asian volume under FOB routinely achieves 25–40% lower ocean rates than the individual CIF rates previously paid by suppliers. The transition requires process discipline (booking, documentation, insurance) but the leverage created by volume consolidation almost always pays for the additional management cost many times over.

Rate management in a volatile market

Ocean freight indices can swing 200–300% over a 2–3 year cycle. Annual contracts with semi-annual review clauses protect both capacity and price. Spot-only strategies expose the shipper to extreme peaks. Systematic invoice auditing recovers 1–3% of annual freight from incorrect surcharges (BAF, GRI, PSS, etc.).

Intermediate vs Advanced

Intermediate: applies the correct Incoterm and verifies shipping documents match the contract.**Advanced**: designs the Incoterm strategy by category and mode, leads CIF->FOB transitions, and runs structured rate benchmarking against market indices.

BEST PRACTICES

Treat the CIF->FOB transition as a formal procurement project

It requires process design, insurance coverage, documentation discipline and change management with suppliers. Done properly it is one of the highest-ROI logistics initiatives available.

The Incoterm on the Bill of Lading is the one with legal effect

It is common for the purchase contract to say FOB while the BL says CIF because the supplier booked the freight for convenience. Always verify the shipping document.

Audit every carrier invoice systematically

Incorrect application of surcharges is routine. Systematic auditing recovers 1–3% of annual freight spend.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — CIF to FOB transition, $12M annual Asian imports — Supplier CIF rates were 37% above the consolidated market rate available to the buyer.

Component	Current CIF	Own FOB
Ocean freight rate	$0.38/kg	$0.24/kg
Management + insurance	—	$0.04/kg
Net cost	$0.38/kg	$0.28/kg

Result: Net savings of $0.10/kg × 12M kg = $1.2M per year. Consolidating volume under one freight forwarder created the leverage required for preferred carrier rates.

WHAT YOU WOULD USE

Rate Management & Benchmarking

E2open / Uber Freight (formerly Transplace) — Freight rate management platforms with market benchmarking.

Freightos / Xeneta — Real-time ocean and air rate intelligence for contract vs spot comparison.

HOW IT IS MEASURED

Incoterm Compliance %

(Transactions with correct Incoterm per policy / Total transactions) × 100

Benchmark: 100% of import transactions with the correct Incoterm documented on the BL/AWB

Freight Rate vs Market Benchmark %

(Average freight rate paid / Market index rate) × 100

Benchmark: <105% of the relevant market index (CCFI, SCFI, Baltic Air)

Freight cost is negotiated, contracted and audited — or it is leaked. Put logistics in the Incoterm decision, tender on a cycle, index volatile components and audit every invoice before it is paid.

03 Freight consolidation, LTL vs. FTL & load optimization

Consolidation combines small shipments into fuller, less frequent loads; the LTL vs. FTL decision compares paying per shipment against paying for the whole truck; load optimization fills each trailer by weight and cube without breaking stacking or delivery-sequence rules.

LTL vs FTL is a service and cost decision, not just a price decision

FTL gives direct routing, lower damage risk and faster transit. LTL is cheaper per unit when volume is low but adds handling, transit time and damage exposure. For most Mexican domestic routes the breakeven sits between 10–14 tons. Above that threshold FTL is almost always superior on total cost and service.

Consolidation program design

Map destinations and average daily/weekly volume. High-volume destinations become scheduled FTL. Mid-volume destinations become multi-stop FTL or milk-runs. Low-volume destinations stay LTL or are served by a milk-run network. The biggest benefit is often reduced management complexity, not just the rate.

3D load optimization

Modern TMS load-building engines use 3D bin-packing algorithms to maximize both weight and cube utilization. A 5–8 point improvement in truck fill rate on FTL lanes is common after implementation and directly reduces the number of trucks required.

Intermediate vs Advanced

Intermediate: decides LTL vs FTL for individual shipments and coordinates consolidation when volume allows.**Advanced**: designs the network-wide consolidation strategy, implements load optimization algorithms, and calculates the LTL/FTL breakeven by route and cargo type.

BEST PRACTICES

Build the destination and volume map before designing the consolidation strategy

Without accurate volume and destination data the consolidation design is based on assumptions that usually prove wrong.

Consolidation requires order cut-off discipline

Moving from daily LTL to 2–3 FTL per week forces the commercial team to accept earlier order cut-offs. Without that discipline, consolidation creates late shipments.

Use milk-runs for low-volume destinations

A well-designed milk-run is frequently more efficient than either pure LTL or forced FTL for low-volume points.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — FMCG company, 840 LTL shipments/month — 68% of destinations were candidates for consolidation into FTL.

Destination	Before	After
North DC (3.2 t/day)	12–15 LTL/week	3 FTL/week (-28% cost)
South DC (2.8 t/day)	10–12 LTL/week	2–3 FTL/week (-22% cost)
Regional stores (<0.5 t/week)	LTL	Milk-run (no change)

Result: 840 LTL -> 320 shipments (140 FTL + 180 LTL). Freight cost -21%. Management complexity -62%.

WHAT YOU WOULD USE

Load Optimization

SAP TM — Load Building — 3D load optimization and FTL/LTL decision engine.

Blue Yonder / Ortec — Advanced 3D packing algorithms and multi-order consolidation.

HOW IT IS MEASURED

Truck Fill Rate %

(Actual volume or weight / Truck capacity) × 100

Benchmark: >85% for FTL · >70% for large LTL

LTL to FTL Conversion Rate %

(Volume converted from LTL to FTL / Prior LTL volume) × 100

Benchmark: Each point of conversion typically yields 15–30% savings on that volume

Fill the truck on purpose. Find the LTL–FTL breakeven by lane, set consolidation windows by customer segment, and make trailer utilization a KPI someone owns.

04 International freight: ocean, air & multimodal

International freight moves goods across borders by ocean (FCL/LCL containers), air, or multimodal combinations, coordinated with freight forwarders, ports, customs brokers and the documentation each leg requires.

International freight is the most complex and highest-variability logistics process

It involves the largest number of parties (freight forwarder, ocean/air carrier, customs broker, terminals), the highest documentary complexity, and the greatest rate volatility. Professional management routinely reduces international logistics cost 15–30% versus relying on the supplier or an unmonitored broker.

Critical documentation and why it fails

Bill of Lading (title of ownership for ocean), Air Waybill, Commercial Invoice, Packing List, and Certificate of Origin (essential for T-MEC benefits). Incorrect or incomplete documentation is the single most frequent cause of customs holds and demurrage.

Visibility and container tracking

Real-time container tracking (vessel position + terminal events) converts pipeline inventory from a black box into a manageable asset. Planning teams that can see delays early can trigger alternative supply or expedite decisions before the stockout occurs.

Intermediate vs Advanced

Intermediate: coordinates with the freight forwarder on assigned shipments and verifies documentation before clearance.**Advanced**: manages relationships with ocean carriers and freight forwarders as strategic partners, designs the consolidation strategy, and owns the international freight P&L.

BEST PRACTICES

Consolidate international volume with the fewest possible freight forwarders

Negotiating power with ocean carriers and airlines is a direct function of committed volume.

Prefer annual contracts over pure spot for the majority of volume

Ocean rates are extremely volatile. Annual contracts with review clauses provide both price and capacity certainty.

Implement real-time container tracking from the day of booking

Visibility that arrives only when the container is already late has limited value.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Manufacturing company, $4.8M ocean freight/year — Three different freight forwarders by region with no volume consolidation.

Initiative	Savings	Additional impact
Consolidation to 1 global forwarder	-14% rate	Preferred carrier access
Annual contracts vs pure spot	-8%	Capacity certainty
Real-time container tracking	—	4 potential stockouts mitigated
Freight invoice audit	$84K recovered	Process discipline

Result: Total savings $720K/year (15%). The largest single lever was consolidating volume under one freight forwarder.

WHAT YOU WOULD USE

International Freight Platforms

Flexport — Digital freight forwarding with end-to-end visibility.

project44 / Descartes — Real-time ocean and air visibility with predictive ETAs.

HOW IT IS MEASURED

Ocean Freight Cost per TEU

Total ocean freight cost / Number of TEUs

Benchmark: Compare against the relevant CCFI corridor index

Air Freight Cost per kg

Total air freight cost / Kilograms shipped

Benchmark: Compare against Baltic Air Freight Index for the corridor

Manage international freight by its variability. Pick the mode by lane economics, plan with observed transit times, qualify forwarders on reliability and get documentation right the first time.

05 Freight cost allocation & logistics P&L; management

Freight cost allocation assigns transport spend to the products, customers and channels that caused it, so the logistics P&L shows where money is made and lost — the basis of cost-to-serve analysis.

The logistics P&L makes the real cost of movement visible

Transportation cost is not a single global number. It is the result of thousands of decisions about mode, carrier, order size, destination and urgency. A logistics P&L broken down by business line, channel, customer and product family is the prerequisite for managing logistics efficiency and for making pricing decisions based on true cost-to-serve.

Cost-to-Serve at customer and order level

Many customers that look profitable at gross margin are unprofitable once full logistics cost is allocated — especially customers with small, frequent, multi-SKU orders or remote delivery points. Cost-to-Serve analysis is the foundation of differentiated pricing, minimum order policies and delivery charges.

Allocation methodology matters

Average cost allocation hides the truth. High-cost customers subsidize low-cost customers. Activity-based or shipment-level allocation reveals which channels and customers actually destroy margin.

Intermediate vs Advanced

Intermediate: reports freight cost by mode and route and identifies deviations versus budget.**Advanced**: designs the full logistics P&L, calculates Cost-to-Serve by customer and order, and uses the results to redesign commercial policies.

BEST PRACTICES

Build the logistics P&L with real allocation, not averages

An “average” Cost-to-Serve of 7% hides that Customer A is at 2.8% and Customer B is at 18.4%.

Cost-to-Serve is the foundation of differentiated logistics pricing

Minimum order quantities, delivery charges by zone, and urgency premiums become rational only when true cost is known.

Integrate the logistics P&L with the financial P&L

When logistics cost is only visible inside the logistics department it never influences commercial or pricing decisions.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — B2B distribution company, 240 active customers — First-time Cost-to-Serve analysis by customer.

Customer Segment	Revenue Share	Average CTS
Top 20 customers	82%	4.2% of revenue
Mid-tier	14%	9.8% of revenue

Customer Segment	Revenue Share	Average CTS
Long tail (80 customers)	4%	22.4% of revenue

Result: 34 long-tail customers had CTS >30% and were structurally unprofitable. Actions: minimum order $800, differentiated delivery charges, and migration of 28 low-frequency customers to self-service. Impact: -$280K annual outbound freight and +1.8 pp net margin.

WHAT YOU WOULD USE

Logistics Costing

SAP S/4HANA — Transportation Costing & Profitability Analysis — Freight cost allocation to delivery orders and logistics P&L by channel, customer and SKU.

Power BI — Logistics Cost Dashboard — Budget vs actual, Cost-to-Serve by customer and channel.

HOW IT IS MEASURED

Logistics Cost as % of Revenue

Total logistics cost / Revenue × 100

Benchmark: 3–6% B2B distribution · 5–8% retail · 8–12% last-mile e-commerce

Cost to Serve per Order

(Freight + processing + packaging + returns) / Number of orders

Benchmark: $15–45/order efficient B2B · >$80/order usually signals orders that are too small

If freight is overhead, nobody manages it. Capture it per shipment, allocate it by causation, and put cost-to-serve in front of the people who set prices and service terms.

Carrier Management & TMS

The operational execution layer — the systems, the carriers and the routes that physically move product and generate or destroy the OTIF promise made to the customer.

01 Carrier selection, qualification & performance management

Carrier management is the cycle of selecting, qualifying, contracting and measuring the transport companies that move your freight — and allocating volume to the ones that perform.

The carrier is the most critical supplier in the physical network

The carrier has physical custody of the product in the final leg. Its failure directly impacts the customer experience and the OTIF promise. Professional carrier management requires the same rigor applied to any strategic supplier: objective selection criteria, systematic measurement, volume-based incentives, and structured relationship management with the most important providers.

Selection and qualification criteria

(1) Safety and regulatory compliance (valid SICT permits, insurance). (2) Network coverage and capacity. (3) Historical OTIF — the single most important metric. (4) Competitive rate. (5) Technology and visibility capability (GPS, EDI/API integration). (6) Financial stability of the carrier.

The carrier scorecard and volume allocation

Core metrics: OTIF %, Damage Rate %, Claims Rate %, Documentation Compliance %, and actual rate vs contracted rate. Monthly review with primary carriers, quarterly with secondary. The decisive practice is linking scorecard performance directly to volume allocation decisions so that poor performers lose volume and strong performers gain it.

Intermediate vs Advanced

Intermediate: evaluates assigned carrier performance and coordinates claims resolution.**Advanced**: designs the carrier management program, leads the annual transportation bid, and owns the relationship with strategic carriers through formal QBRs.

BEST PRACTICES

Consolidate the carrier base — more volume per carrier creates leverage

A shipper with 30 carriers has little influence over any of them. Concentrating 90% of volume in 5–8 selected carriers changes the power balance.

Link the scorecard directly to volume allocation

When carriers know that low OTIF results in less volume, the incentive structure becomes self-reinforcing.

SICT certification and valid insurance are non-negotiable

These are prerequisites, not nice-to-haves. Verification must be part of the formal qualification process.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Distribution company with 28 active carriers — Eight carriers generated 78% of shipments with highly variable OTIF.

Segment	Carriers	Strategy
Primary (OTIF >96%)	5	Long-term agreements + quarterly QBR
Mid-performance (90–95%)	8	Formal improvement plan + monthly review
Low performance (<90%)	6	Volume reduction + replacement within 90 days
Low frequency	9	Consolidate into 3 qualified carriers

Result: Carrier base reduced from 28 to 14 in six months. Average OTIF rose from 88.4% to 94.2%. Greater volume concentration created leverage for better rates and tighter SLAs.

WHAT YOU WOULD USE

Carrier Management

SAP TM — Carrier Management — Qualification, scorecard and contract management inside SAP TM.

Power BI — Carrier Scorecard — OTIF by carrier and route, damage rate, and performance versus target.

HOW IT IS MEASURED

Carrier OTIF %

(Complete and on-time deliveries / Total deliveries) × 100

Benchmark: >96% primary carriers · >92% secondary carriers

Cargo Damage Rate %

(Shipments with damage claim / Total shipments) × 100

Benchmark: <0.5% for qualified carriers · >1.5% requires immediate action

The carrier is your service in the customer's eyes. Qualify hard, consolidate the base, score monthly and move volume toward the carriers that keep your promises.

02 TMS implementation & transportation planning

A Transportation Management System (TMS) plans, executes and settles freight: it turns orders into loads, selects and books carriers, tracks shipments and audits invoices, integrated with the ERP and WMS.

A well-implemented TMS reduces freight cost 8–15% and frees planning capacity

Core value stack: (1) Order management and ERP integration, (2) Load planning and route optimization, (3) Automated carrier selection and booking, (4) Real-time shipment tracking, (5) Automated freight audit and invoice matching. Without clean master data and process discipline the TMS simply automates the existing chaos.

Integration is the prerequisite of value

ERP -> TMS (delivery orders become shipments), WMS -> TMS (dispatch confirmation), TMS -> ERP (delivery confirmation and freight cost). A TMS disconnected from the ERP creates re-keying, lost traceability and limited audit capability.

Phased implementation reduces risk

Start with carrier booking automation (fastest visible ROI and lowest risk). Then load optimization. Then freight audit. Attempting to implement everything simultaneously is the most common cause of cost overruns and low adoption.

Intermediate vs Advanced

Intermediate: uses the TMS for day-to-day operations and coordinates with warehouse and carriers.**Advanced**: designs the TMS architecture, leads implementation, defines optimization parameters and process KPIs, and drives continuous improvement of the system.

BEST PRACTICES

ERP–TMS integration is non-negotiable

Without it the TMS becomes another disconnected system that creates work instead of removing it.

Define optimization parameters before configuration

Changing parameters during implementation is the most frequent cause of delays and cost overruns.

Treat change management as 50% of the project

A TMS that is used only as a recording system instead of an optimization engine delivers a fraction of its potential value.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — FMCG company, 1,200 transport orders/month — Logistics team spent ~70% of time on manual coordination.

Module	Timeline	Impact
Order Management + auto booking	3 months	Booking time 45 min -> 8 min (-82%)
Load Planning + route optimization	4 months	Truck fill rate 74% -> 88% · Freight -9%
Real-time tracking	4 months	In-transit visibility 0% -> 94%
Automated freight audit	5 months	$84K incorrect charges recovered in first 6 months

Result: Freight cost -11.4% ($480K/year). Team capacity freed allowed reallocation of 3 FTEs to network optimization projects.

WHAT YOU WOULD USE

TMS Platforms

SAP TM — Native integration with S/4HANA, strong for complex multi-modal environments.

Blue Yonder TMS / Oracle OTM / Manhattan — Best-of-breed options with advanced optimization engines.

HOW IT IS MEASURED

TMS Automation Rate %

(Orders processed without manual intervention / Total orders) × 100

Benchmark: >80% of orders processed automatically

TMS-Driven Savings %

(Post-TMS freight cost - Pre-TMS) / Pre-TMS × 100

Benchmark: 8–15% freight reduction in the first 12 months is typical for a well-executed implementation

Fix the process, then automate it. Document the rules, clean the data, integrate with ERP and WMS, and roll out in phases that each pay for themselves.

03 Route optimization & territory design

Route optimization assigns stops to vehicles and sequences them to minimize distance, time or cost under real constraints — time windows, capacity, driver hours, vehicle restrictions. Territory design divides the service area into balanced zones that make daily routing stable.

Route design is a continuous optimization problem, not a one-time exercise

Fixed routes provide stability and driver familiarity. Dynamic routes (solved daily by a VRP engine) provide higher utilization and better service when demand is variable. Most mature operations use a hybrid: fixed backbone routes for high-volume customers and dynamic optimization for the variable tail.

Mexican urban constraints change the problem

Circulation restrictions (Hoy No Circula, local truck bans, time windows, height and weight limits) must be modeled explicitly. A route that looks optimal on a pure distance or time basis can be illegal or impractical in Mexico City, Guadalajara or Monterrey.

Territory design precedes daily routing

Poor territory design forces the daily optimizer to compensate for structural imbalance. Balanced territories by workload, geographic contiguity and customer service requirements produce more stable and efficient daily routes.

Intermediate vs Advanced

Intermediate: adjusts routes within existing territories and handles exceptions.**Advanced**: redesigns territories periodically, configures the VRP engine with realistic constraints, and continuously improves the optimization model.

BEST PRACTICES

Redesign territories before optimizing daily routes

Daily optimization cannot fully compensate for poorly balanced territories.

Model real Mexican urban constraints explicitly

Time windows, truck bans, height restrictions and security considerations must be in the model or the optimized routes will be rejected by operations.

Review territory design at least annually

Customer density and order patterns drift. Static territories become inefficient within 12–18 months in growing markets.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Distribution company in central Mexico — Fixed routes designed three years earlier no longer matched demand density.

Metric	Before	After territory redesign + dynamic routing
Average stops per route	14.2	18.7
Km per delivery	18.4 km	13.1 km (-29%)
On-time delivery	91%	96%
Fleet utilization	71%	86%

Result: Two vehicles removed from the fleet while improving service. Annual savings approximately $1.1M in operating cost.

WHAT YOU WOULD USE

Route Optimization

SAP TM / Blue Yonder / Ortec — Mature VRP engines with support for time windows, vehicle constraints and multi-day planning.

Google OR-Tools / specialized last-mile solvers — Used by companies that build custom optimization layers on top of their TMS.

HOW IT IS MEASURED

Km per Delivery / Cost per Stop

Total km or total route cost / Number of deliveries

Benchmark: Declining trend is the goal; absolute number depends on geography and density

Stops per Route / Route Utilization %

Average number of productive stops and % of available capacity used

Benchmark: Higher is better until service or driver hours constraints are hit

Optimize inside reality. Fix territories first, model the real urban constraints, mix fixed and dynamic routing, and keep tuning the model with what drivers actually do.

04 Real-time shipment visibility & supply chain control tower

Shipment visibility tracks freight in real time across carriers and modes; a logistics control tower adds alert thresholds, owners and escalation rules so exceptions are detected and resolved before the customer is affected.

Visibility without action is just expensive tracking

A logistics control tower combines multi-source data (TMS, carrier GPS, ocean/air events, weather, traffic) with defined alert thresholds and escalation protocols so that exceptions are detected and resolved before the customer is impacted. The goal is

proactive intervention, not reactive firefighting.

Alert design determines whether the tower creates value or noise

Too many alerts produce alert fatigue and ignored signals. Too few alerts mean problems are discovered late. Effective towers use tiered thresholds (warning -> critical) and route alerts to the role that can actually act on them.

Closed-loop exception management

Detection -> ownership assignment -> action -> customer communication (when required) -> root-cause capture -> process improvement. Towers that stop at detection create visibility without accountability.

Intermediate vs Advanced

Intermediate: monitors active shipments and escalates exceptions according to protocol.**Advanced**: designs the control tower architecture, defines alert logic and escalation paths, and uses the data to drive carrier and process improvement.

BEST PRACTICES

Design alerts for action, not for information

Every alert should have a clear owner and a defined next step. Alerts that no one is expected to act on should not exist.

Close the loop into carrier and process improvement

Repeated exceptions with the same root cause must feed the carrier scorecard and internal process reviews.

Proactive customer communication is a core tower output

A customer who is told about a delay before they have to ask experiences a different level of service.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Mid-sized manufacturer implementing a control tower — Previously learned about delays only when the customer called.

Metric	Before	After 9 months
% of delays detected before customer contact	12%	81%
Average time from delay detection to customer notification	N/A (reactive)	4.2 hours
Customer complaints related to late delivery	Baseline	-34%
Expedite freight triggered by late discovery	Baseline	-27%

Result: The largest gain was not operational cost reduction but the shift from reactive to proactive customer communication, which measurably improved perceived reliability.

WHAT YOU WOULD USE

Visibility & Control Tower

project44 / FourKites / Shippeo — Leading multimodal visibility platforms with predictive ETAs and exception management.

SAP Logistics Control Tower / Blue Yonder Luminate — Control tower capabilities integrated with broader planning suites.

HOW IT IS MEASURED

Proactive Detection Rate %

(Delays detected by the tower before customer contact / Total delays) × 100

Benchmark: >75% is a solid operating target

Mean Time to Detect / Mean Time to Resolve

Average hours from event occurrence to detection and from detection to resolution

Benchmark: Track both; detection without resolution creates little value

Build the tower around decisions, not screens. Unify the data, give every alert an owner, warn customers early and turn recurring exceptions into fixes upstream.

05 Fleet management: own fleet vs. 3PL decision

The own-fleet vs. 3PL decision compares running private or dedicated vehicles against buying capacity from carriers and 3PLs — on full cost, service control, flexibility and risk — and usually ends in a hybrid.

The own-fleet vs 3PL decision is a total-cost and strategic-control decision

Own fleet provides maximum control over service, branding, security and capacity. 3PL provides flexibility, lower fixed cost and access to broader networks. The correct answer is rarely 100% one or the other. Most mature operations run a hybrid model: own or dedicated fleet for the stable, high-volume core and 3PL capacity for peaks, geographic coverage and specialized needs.

Full TCO of an owned truck

Depreciation or lease, fuel, driver wages and benefits, maintenance, insurance, tires, permits (including NOM-087 related costs where applicable), tolls, and the management overhead of running a fleet. Many companies understate the true cost of ownership by omitting management time and downtime.

Breakeven and hybrid design

Calculate the utilization level at which own fleet becomes cheaper than 3PL on a pure cost basis, then adjust for the value of control, service differentiation and capacity assurance. The hybrid model usually emerges as the optimal solution once both cost and strategic factors are considered.

Intermediate vs Advanced

Intermediate: manages day-to-day fleet or 3PL operations within the existing model.**Advanced**: runs the full make-vs-buy analysis, designs the hybrid model, and periodically re-evaluates the boundary between own and outsourced capacity.

BEST PRACTICES

Calculate full TCO of ownership before deciding

Omitting management overhead, downtime and residual value risk systematically biases the analysis toward ownership.

Default to hybrid unless there is a strong strategic reason for pure models

Pure own fleet is rigid. Pure 3PL can leave the shipper exposed on capacity and service. Hybrid captures the advantages of both.

Re-evaluate the boundary every 2–3 years

Freight market conditions, volume patterns and internal capabilities change. The optimal split is not static.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Regional distributor evaluating fleet strategy — Operated a pure 3PL model with rising rates and declining capacity commitment during peaks.

Option	Annual Cost	Service & Control
100% 3PL (current)	Baseline	Low control, capacity risk in peak
100% own fleet	+18%	High control, high fixed cost, low flexibility
Hybrid (60% dedicated + 40% 3PL)	-4% vs current	Core volume protected, peak flexibility retained

Result: Hybrid model implemented. Core lanes moved to dedicated fleet with service-level guarantees. Peak and long-tail volume remained with 3PLs. Overall cost slightly lower and peak-season OTIF improved 5 points.

WHAT YOU WOULD USE

Fleet & Dedicated Fleet Management

SAP TM / fleet modules in major TMS — Support for private and dedicated fleet planning alongside common carrier.

Specialized fleet management systems — Used when own fleet is large enough to justify dedicated maintenance, fuel and compliance management tools.

HOW IT IS MEASURED

Cost per Km / Cost per Delivery (own vs 3PL)

Fully loaded cost comparison on comparable lanes

Benchmark: Must include all ownership costs; partial costing leads to wrong decisions

Fleet Utilization % / Dedicated Fleet OTIF

Utilization of owned or dedicated assets and their service performance

Benchmark: Own/dedicated fleet should outperform 3PL on OTIF if the control premium is to be justified

Decide on full cost and on control, then go hybrid. Count every ownership cost, price the market honestly, value control explicitly, and revisit the split on a schedule.

Cold Chain & Specialized Logistics

The specialized logistics layer — segments where standard rules do not apply and where compliance, engineering and execution precision determine whether value is preserved or destroyed in transit.

01 Cold chain design, temperature monitoring & compliance

Cold chain is the controlled-temperature system — packaging, equipment, monitoring, procedures and qualified partners — that keeps pharmaceuticals, biologics and perishable foods within their specified range from origin to delivery, with documented evidence that it did.

Cold chain is a compliance and product-integrity system, not just refrigerated transport

For pharmaceuticals, biologics, certain foods and chemicals, temperature excursions can destroy product value or create patient safety risk. In Mexico, COFEPRIS requirements make continuous temperature logging and documented investigation of excursions a regulatory obligation, not an optional best practice.

Monitoring architecture

Modern practice uses IoT data loggers with real-time or near-real-time transmission and pre-limit alert thresholds (warning before the regulatory limit is breached). Passive-only loggers that are read at destination detect problems too late to prevent loss. The monitoring system must cover the entire chain: warehouse, loading, transit, and receiving.

Excursion investigation protocol

Every confirmed excursion requires a documented investigation: what happened, product impact assessment, disposition decision (release, quarantine, reject), and corrective/preventive action. Without a formal protocol, organizations either over-reject (destroying good product) or under-investigate (releasing compromised product).

Intermediate vs Advanced

Intermediate: executes monitoring and follows the excursion protocol for assigned shipments.**Advanced**: designs the end-to-end cold chain (packaging, mode, monitoring, SOPs), qualifies providers, and owns the compliance system with COFEPRIS and customer audits.

BEST PRACTICES

Move from passive to active monitoring for high-value or high-risk products

Passive loggers only tell you that something went wrong after the damage is done.

Treat every excursion as a formal quality event

Documented investigation, impact assessment and CAPA are required both for compliance and for learning.

Qualify the entire chain, not just the carrier

Warehouse docks, staging areas, and receiving processes are frequent sources of excursions that never appear in transit data.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case — Pharmaceutical distributor, Mexico** — Previously used passive loggers read only at destination. Several excursions were discovered after product had already been received into inventory.

Element	Before	After
Monitoring	Passive loggers	IoT loggers with pre-limit alerts
Time to detect excursion	At destination (often 24–72 h later)	Average 47 minutes from breach
Product loss from excursions	Baseline	-61% in 12 months
Audit findings related to temperature control	Recurring	Zero major findings in subsequent COFEPRIS inspection

Result: Real-time visibility allowed intervention (re-icing, re-routing, or accelerated delivery) before product was compromised. The compliance posture improved materially.

WHAT YOU WOULD USE

Cold Chain Monitoring

Sensitech / Controlant / Berlinger / Roambee — Leading IoT temperature monitoring platforms with real-time alerts and compliance reporting.

Integration with QMS / Track & Trace systems — Excursion data should flow into the quality management system for formal investigation and disposition.

HOW IT IS MEASURED

Temperature Excursion Rate %

(Shipments with confirmed excursion / Total cold-chain shipments) × 100

Benchmark: <0.5% for mature pharma cold chains · higher rates require root-cause program

Mean Time to Detect Excursion

Average time from temperature limit breach to system alert / human awareness

Benchmark: <1 hour for real-time systems · days for passive loggers

Design for evidence, not just for cold. Qualify every link, alert before the limit, and treat every excursion as a quality event with a documented decision.

02 Hazmat logistics: ADR/DOT compliance & dangerous goods

Hazmat logistics is the regulated movement of dangerous goods: correct UN classification, approved packaging and labeling, required documentation, qualified carriers and trained personnel, under national rules (NOM-002-SCT and related in Mexico, DOT in the US) and international codes (ADR, IMDG, IATA DGR).

Hazmat is a regulated activity with personal and corporate liability

Misclassification, incorrect packaging, missing documentation or untrained personnel can result in fines, shipment rejection, accidents and criminal liability. In Mexico the primary references are NOM-002-SCT and related standards; internationally ADR (road), IMDG (sea) and IATA DGR (air) apply. Compliance is not optional and cannot be fully delegated to the carrier.

Core compliance elements

(1) Correct UN classification and proper shipping name. (2) Approved packaging and marking/labeling. (3) Required documentation (Dangerous Goods Declaration, emergency response information). (4) Carrier qualification for the specific hazard class. (5) Training of all personnel who prepare, offer or transport dangerous goods (recurrent training is mandatory).

Carrier and mode restrictions

Not every carrier is authorized for every hazard class. Some materials are forbidden on passenger aircraft or in certain tunnels. Mode and carrier selection for hazmat must start from regulatory permission, not from cost or transit time.

Intermediate vs Advanced

Intermediate: prepares shipments according to approved procedures and current classifications.**Advanced**: owns the dangerous goods compliance program, maintains classifications, qualifies carriers, designs training, and manages regulatory interactions and audits.

BEST PRACTICES

Maintain a single, controlled source of truth for classifications

Multiple spreadsheets with divergent UN numbers are a common and dangerous failure mode.

Treat training as a formal authorization, not a one-time event

Only trained and currently certified personnel should be allowed to prepare or offer dangerous goods shipments.

Qualify carriers specifically for the hazard classes you ship

General freight authority does not equal dangerous goods authority for every class.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Chemical distributor with recurring documentation rejections — Approximately 4% of hazmat shipments were delayed or rejected at origin or destination due to documentation or labeling errors.

Root Cause	Share of Rejections	Corrective Action
Incorrect or outdated UN classification	31%	Centralized classification master + annual review
Missing or incomplete DG Declaration	27%	System-enforced document checklist before release
Labeling / marking errors	22%	Standardized label kits + photo verification
Untrained personnel preparing shipments	20%	Mandatory recurrent training + authorization matrix

Result: Rejection rate fell from 4% to 0.6% within nine months. More importantly, the company passed a subsequent SCT-related inspection without major findings.

WHAT YOU WOULD USE

Dangerous Goods Compliance

Labelmaster / specialized dangerous-goods software — Specialized dangerous goods software for classification, documentation and labeling.

TMS modules with DG checks — Leading TMS platforms can enforce documentation and carrier qualification rules before shipment release.

HOW IT IS MEASURED

Hazmat Documentation Accuracy %

(Shipments with complete and correct DG documentation / Total hazmat shipments) × 100

Benchmark: >=99.5% · anything lower indicates systemic process failure

Training Compliance %

(Personnel with current required DG training / Personnel who handle DG) × 100

Benchmark: 100% · expired training is a regulatory violation

Make non-compliance impossible to ship. One classification master, system-enforced documents, carriers qualified per class, and trained, authorized people.

03 Oversize, project cargo & specialized equipment logistics

Project cargo covers oversize, overweight and high-value moves — industrial equipment, transformers, turbines, structures — that need route surveys, special permits, escorts, utility coordination and specialized equipment.

Project cargo is a planning and permit problem before it is a transport problem

Oversize and heavy-lift moves require route surveys, multiple permits (SICT, state, municipal), police escorts, utility coordination (lifting lines), and specialized equipment. Lead times for permits in Mexico can range from days to several weeks depending on the corridor and the dimensions/weight. Starting the transport process without the permit path defined is the most common cause of costly delays.

Route survey is non-negotiable

Bridge capacities, turning radii, overhead clearances, road gradients and seasonal restrictions must be verified on the actual route. Desktop planning is insufficient for truly exceptional loads. Specialized carriers that already know the corridors and have existing relationships with authorities significantly reduce both risk and lead time.

Permit timeline management

Build the critical path backward from the required on-site date. Include contingency for permit delays, weather windows and escort availability. Last-minute project cargo is almost always expensive and high-risk.

Intermediate vs Advanced

Intermediate: coordinates with specialized carriers and tracks permit status for assigned moves.**Advanced**: designs the project cargo process, selects and qualifies specialized carriers, manages the permit critical path, and owns the risk assessment for exceptional loads.

BEST PRACTICES

Never skip the physical route survey for truly exceptional loads

Desktop tools miss temporary restrictions, new construction and local conditions.

Start the permit critical path as soon as dimensions and weight are known

Permit lead time is usually the longest item on the schedule.

Use specialized carriers that already own the corridor relationships

Their experience and existing authority relationships reduce both risk and cycle time.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Industrial equipment move, 4.2 m wide × 5.1 m high, 68 tons — Required delivery to a plant in Guanajuato with a fixed shutdown window.

Activity	Planned Lead Time	Actual
Route survey	10 days	12 days (one bridge required reinforcement assessment)
SICT + state + municipal permits	18 days	23 days
Utility coordination (line lifting)	7 days	7 days
Execution window	3 nights	3 nights (completed inside shutdown)

Result: Because the critical path was started early and a specialized carrier with corridor experience was used, the load arrived inside the shutdown window. A competitor that started later missed their window and incurred significant demurrage and schedule penalties.

WHAT YOU WOULD USE

Project Cargo Planning

Specialized project cargo / heavy-lift planning tools — Used by heavy-lift carriers and project logistics providers for route analysis and permit tracking.

Project management + document control systems — Critical for managing the multi-party permit and coordination timeline.

HOW IT IS MEASURED

Permit Cycle Time vs Plan

Actual calendar days to obtain all required permits versus the planned critical path

Benchmark: Track variance; chronic underestimation indicates process or carrier issues

On-Time Project Cargo Delivery %

(Project cargo moves delivered inside the agreed window / Total project cargo moves) × 100

Benchmark: >=95% for a mature process with adequate planning lead time

Start with permits and surveys, not with the truck. Freeze specs early, verify the route physically, plan backward from the site date and hire carriers who already know the corridor.

04 Packaging & unitization for transportation

Transport packaging and unitization design the protective package, the case, the pallet pattern and the load so that product survives the distribution environment while using as much of the vehicle's cube as possible.

Transport packaging is a total-cost decision that balances protection, space and handling

Over-packaging wastes cube and increases freight cost. Under-packaging increases damage, claims and customer dissatisfaction. The correct design is validated by testing (ASTM D4169 or equivalent) and evaluated on total cost: packaging material + freight impact + damage cost + handling cost.

Unitization and pallet configuration

Pallet patterns, stack height, and interlocking determine both stability and vehicle fill rate. A 5–8% improvement in cube utilization on high-volume lanes is frequently achievable through better unitization without changing primary packaging.

Testing and validation

Laboratory testing to recognized protocols (ASTM D4169, ISTA) provides objective evidence that the packaging system can survive the expected distribution environment. For new products or new lanes this testing is far cheaper than learning through field damage.

Intermediate vs Advanced

Intermediate: applies approved packaging specifications and reports damage trends.**Advanced**: designs packaging systems using TCO logic, commissions testing, optimizes unitization, and continuously reduces damage while protecting or improving cube utilization.

BEST PRACTICES

Test before you scale

Field damage is a very expensive way to discover that packaging is inadequate for the distribution environment.

Evaluate packaging changes on total cost, not material cost alone

A cheaper box that increases damage or reduces cube utilization is usually more expensive overall.

Optimize unitization as aggressively as primary packaging

Pallet patterns and stack heights are frequently left unchanged for years even as product dimensions evolve.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Consumer durables manufacturer — Damage rate on a key product family was 2.8% with frequent “crushed corner” claims.

Initiative	Impact on Damage	Impact on Cube / Cost
Reinforced corner posts + better pallet interlocking	Damage -41%	Neutral on cube
Optimized pallet pattern (same protection)	Neutral	+6% units per truck
Elimination of unnecessary void fill	Neutral	-3% packaging material cost

Result: Combined annual benefit approximately $1.4M from reduced damage claims plus freight and material savings. Payback on packaging engineering work was under four months.

WHAT YOU WOULD USE

Packaging & Load Optimization

Cape Pack / TOPS / specialized packaging design tools — Pallet pattern and cube optimization software.

TMS load-building modules — 3D load optimization that respects packaging and stacking constraints.

HOW IT IS MEASURED

Damage Rate % (transport-related)

(Units or shipments with transport damage / Total units or shipments) × 100

Benchmark: Depends on product; track trend and cost of damage, not only frequency

Cube Utilization % / Units per Trailer

Actual product volume or units / Available trailer volume or theoretical maximum

Benchmark: Higher is better provided damage rate does not increase

Engineer packaging on total cost. Baseline damage and cube, test to a standard, optimize unitization and never judge a box by its price alone.

05 E-commerce fulfillment logistics & last mile

E-commerce fulfillment and last mile cover how online orders are picked, packed and delivered to homes, lockers or pickup points — from own DCs, 3PLs, marketplaces or stores — at a cost and speed the customer accepts.

Last-mile cost is the defining economic constraint of e-commerce logistics

Last-mile frequently represents 40–50% of total e-commerce fulfillment cost. The economics are driven by drop density, failed delivery rate, return rate and customer expectation of speed. Models that ignore these variables systematically understate true cost-to-serve.

Fulfillment model choices

D2C from own DC, 3PL fulfillment, marketplace fulfillment (FBA-style), and Ship-from-Store each have different cost structures, speed profiles and control characteristics. Most scaled players operate a hybrid and dynamically allocate orders to the node that best balances cost and promise.

Density and network design

Out-of-home delivery (lockers, collection points) and micro-fulfillment nodes improve density and reduce failed deliveries. In Mexican urban environments, security, access restrictions and traffic make pure home delivery particularly expensive; out-of-home solutions often improve both cost and success rate.

Intermediate vs Advanced

Intermediate: manages day-to-day last-mile operations and carrier performance within the existing model.**Advanced**: designs the multi-node fulfillment network, optimizes order allocation logic, and continuously improves density and cost-to-serve.

BEST PRACTICES

Treat density as a first-class design variable

Every initiative that increases drops per km or reduces failed attempts improves the fundamental economics.

Design for hybrid fulfillment from the start

Pure single-node models rarely remain optimal as volume and geographic coverage grow.

Integrate returns into the same network

Out-of-home points that also accept returns improve both customer experience and reverse-logistics cost.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Mid-sized e-commerce player in Mexico — Pure home-delivery model with rising last-mile cost and 18% failed-delivery rate on first attempt.

Initiative	Impact
Introduction of locker + collection-point network in top 8 cities	First-attempt success +11 pp · Last-mile cost -14% on covered volume
Dynamic allocation between own DC and 3PL nodes	Average delivery distance -18%
Improved address validation + customer communication	Failed deliveries -23%

Result: Blended last-mile cost per order reduced 17% while average delivery speed improved. The locker network also reduced reverse-logistics cost because returns could be dropped at the same points.

WHAT YOU WOULD USE

E-commerce & Last-Mile Platforms

Delivery platforms + locker networks (local and regional providers) — Critical for out-of-home density in Mexican cities.

OMS + TMS with multi-node allocation — Order management systems that can intelligently choose fulfillment node and last-mile method.

HOW IT IS MEASURED

Last-Mile Cost per Order

Total last-mile cost / Number of delivered orders

Benchmark: Highly dependent on density and model; track trend and compare across nodes

First-Attempt Delivery Success %

(Orders delivered on first attempt / Total delivery attempts) × 100

Benchmark: >90% is a solid target in urban environments with good address data

Win the last mile on density and first-attempt success. Measure cost per delivered order, add out-of-home options, fix addresses at checkout and route each order to the right node.

Network Design & 3PL Management

The strategic layer of logistics — decisions that determine the structural cost and service capability of the network for the next 3–5 years, and the partnerships that execute it day to day.

01 Logistics network design: facility location & flow optimization

Logistics network design decides how many facilities to operate, where, with what role, and how product flows between plants, DCs and customers — using optimization models that balance transport, facility and inventory cost against service.

Network design determines structural cost and service capability for years

Facility location and flow decisions lock in the majority of logistics cost and the achievable service levels. Adding or removing a DC, changing the primary flow paths, or altering the split between central and regional inventory creates multi-year consequences. These decisions should be made with formal Supply Chain Network Design (SCND) models, not by incremental local optimization.

The fundamental trade-off: nodes vs cost vs service

More nodes generally improve service (shorter distance to customer) and can reduce outbound transportation cost, but they increase facility cost, inventory (less risk pooling), and complexity. Fewer nodes do the opposite. The optimal number and location of nodes depends on service requirements, demand density, product characteristics and transportation rates.

Risk pooling and inventory implications

Centralizing inventory reduces safety stock through risk pooling. Decentralizing improves responsiveness but multiplies inventory. Network design and inventory strategy must be optimized together; treating them sequentially produces suboptimal results.

Intermediate vs Advanced

Intermediate: understands the current network logic and contributes data to design studies.**Advanced**: leads network design projects, builds or directs SCND models, quantifies service–cost–inventory trade-offs, and owns the transition plan from current to future state.

BEST PRACTICES

Optimize network and inventory together

Sequential optimization (first network, then inventory) systematically leaves money on the table.

Always include a realistic transition plan

The optimal future-state network is useless if the company cannot get there without breaking service or incurring prohibitive change costs.

Re-evaluate the network when volume, mix or service strategy changes materially

A network designed five years ago for a different demand footprint is often no longer optimal.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> **Illustrative case — National distributor evaluating a 2-DC vs 4-DC network** — Service target: 95% of demand within one-day ground transit.

Network Option	Outbound Freight	Facility + Inventory Cost	Service (% demand <=1 day)
Current (2 DCs)	Baseline	Baseline	81%
3-DC optimized	-11%	+9%	94%
4-DC optimized	-16%	+21%	97%

Result: The 3-DC design met the service target at the lowest total cost. The 4-DC option improved service only marginally while adding significant fixed and inventory cost. A phased transition plan was executed over 14 months.

WHAT YOU WOULD USE

Network Design

Coupa Supply Chain Design (formerly LLamasoft) / anyLogistix / AIMMS — Leading Supply Chain Network Design platforms.

Blue Yonder / Kinaxis scenario capabilities — Useful for ongoing what-if analysis once the baseline network is established.

HOW IT IS MEASURED

% Demand within Target Transit Time

Share of demand that can be reached within the defined service standard from the current network

Benchmark: Set by commercial strategy; network design should achieve it at minimum total cost

Total Network Logistics Cost (Freight + Facility + Inventory)

Fully loaded structural cost of the network

Benchmark: Primary objective function of network design studies

Design the network on total cost and service, with inventory inside the model. Start from a validated baseline, test scenarios, and plan a transition the business can actually execute.

02 3PL selection, contract design & governance

3PL management covers selecting a third-party logistics provider, structuring the contract — scope, SLAs, pricing, data and exit terms — and governing the relationship with scorecards and regular business reviews.

A 3PL relationship is only as good as the contract and the governance model

Selecting a 3PL on price alone is the most reliable way to create future service and cost problems. The RFP, site visits, reference checks, SLA design, pricing structure, exit clauses and ongoing governance together determine whether the relationship creates value or becomes a source of chronic friction.

Critical contract elements

Clear scope and volume assumptions, measurable SLAs with consequences (credits or penalties), transparent pricing (including how gain-share or inflation is handled), data ownership and system access rights, continuous improvement expectations, and practical exit provisions (transition assistance, data return, non-solicitation).

Active governance prevents performance drift

Quarterly business reviews with scorecards, root-cause analysis of misses, joint improvement projects, and escalation paths keep the relationship aligned. Passive governance that only reacts when service collapses is too late.

Intermediate vs Advanced

Intermediate: manages day-to-day 3PL performance against the existing contract.**Advanced**: designs the selection process, structures the contract and SLAs, and runs the governance cadence that keeps the relationship healthy and improving.

BEST PRACTICES

Never accept vague SLAs

“Best effort” or “industry standard” language is unenforceable and guarantees future disputes.

Design the exit before you sign the entry

The time to negotiate transition assistance and data rights is before the relationship begins, not when it is already failing.

Run governance with the same rigor as an internal department

A 3PL that is not reviewed regularly will optimize for its own convenience.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Manufacturer outsourcing a regional DC to a new 3PL — Previous 3PL relationship had degraded due to weak SLAs and almost no governance.

Element	Previous Contract	New Contract + Governance
SLAs	Vague (“best effort”)	Specific OTIF, accuracy, dock-to-stock with credits
Pricing	Partially opaque	Activity-based + clear gain-share mechanism
Governance	Ad-hoc when problems arose	Monthly ops + quarterly executive QBR
Exit provisions	Weak	Detailed transition assistance and data return

Result: After 12 months the new relationship delivered 4.1 points higher OTIF and a 7% reduction in cost-to-serve versus the previous 3PL, with far fewer escalations.

WHAT YOU WOULD USE

3PL Management

Scorecard & QBR tools (Power BI / Tableau + shared data room) — Most effective governance stacks combine operational data with structured review cadences.

Contract lifecycle / CLM systems — Useful for managing complex multi-year 3PL agreements and amendments.

HOW IT IS MEASURED

3PL SLA Attainment %

(SLAs met / Total measured SLAs) × 100 or weighted scorecard result

Benchmark: Target should be set high enough that misses trigger meaningful discussion

Cost-to-Serve vs Contract Baseline

Actual fully-loaded cost versus the cost assumed in the commercial model

Benchmark: Significant favorable or unfavorable variance requires investigation

Outsource execution, keep control. Specify scope precisely, make SLAs bite, design the exit before signing, and run governance with the same rigor as an internal operation.

03 Logistics KPIs, benchmarking & performance dashboards

The logistics KPI system defines a small set of metrics by level — operational, tactical, strategic — with clear definitions, owners and targets, and compares them with internal history and external benchmarks.

Logistics performance must be measured at three levels

Operational (daily/weekly execution metrics), tactical (monthly service, cost and productivity), and strategic (structural cost, network productivity, contribution to cash and customer experience). A single flat list of KPIs serves no audience well.

Perfect Order Rate as the integrating service metric

Perfect Order Rate (on time × in full × damage-free × correct documentation) is the most complete single measure of logistics service. It forces the organization to manage the whole chain rather than optimizing individual legs in isolation.

Translate logistics metrics into financial language

CFO and general management audiences respond to cost-to-serve, logistics cost as % of revenue, cash impact of inventory in transit, and the financial cost of service failures. Pure operational metrics rarely drive resource decisions at the top of the house.

Intermediate vs Advanced

Intermediate: reports the established KPI set accurately and on time.**Advanced**: designs the KPI framework by audience, builds the dashboards, runs benchmarking, and translates operational results into financial and strategic insight.

BEST PRACTICES

Design KPIs for decisions, not for reporting volume

If a metric does not change a decision or trigger an action, question why it is on the dashboard.

Adopt Perfect Order Rate as the primary service metric

It prevents the classic trap of optimizing on-time delivery while destroying in-full or damage performance.

Always maintain a financial translation layer

Operational excellence that cannot be expressed in cost, cash or customer impact struggles to attract investment.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Company with 40+ logistics reports and no single version of truth — Leadership regularly received conflicting numbers on OTIF and freight cost.

Action	Result
Defined a three-level KPI framework with clear owners	Eliminated competing definitions of OTIF and cost
Built one operational + one executive dashboard	Reduced reporting effort ~40% while improving clarity
Added Perfect Order Rate and Cost-to-Serve	Shifted conversation from activity metrics to outcomes
Instituted quarterly external/internal benchmarking	Identified two structural gaps previously invisible

Result: Within two quarters the leadership conversation moved from debating whose numbers were correct to deciding which gaps to close first.

WHAT YOU WOULD USE

Logistics Performance Management

Power BI / Tableau / Looker — Primary tools for operational and executive logistics dashboards.

Control tower and TMS analytics modules — Provide the granular event data that feeds higher-level KPI systems.

HOW IT IS MEASURED

Perfect Order Rate %

(Orders that are on time, in full, damage-free and correctly documented / Total orders) × 100

Benchmark: World-class operations target mid-to-high 90s; most companies start materially lower

Logistics Cost as % of Revenue / Cost-to-Serve

Total logistics cost relative to revenue and fully-loaded cost per order or per customer

Benchmark: Industry and channel dependent; trend and peer comparison matter most

Fewer metrics, one definition each, tied to decisions. Tier by audience, anchor on perfect order and cost-to-serve, and benchmark to find the gaps worth closing.

04 Cross-border logistics: customs, nearshoring & free trade zones

Cross-border logistics integrates physical movement with trade compliance: tariff classification, rules of origin under the USMCA (T-MEC), programs such as IMMEX, customs brokerage and border crossing operations.

Cross-border logistics is where trade compliance and physical movement intersect

For Mexico-based and Mexico-serving supply chains, T-MEC (USMCA) rules of origin, tariff classification, IMMEX program benefits, and customs processing times are structural determinants of cost and lead time. Treating customs as a pure administrative afterthought destroys value that nearshoring was intended to create.

T-MEC utilization and origin management

Preferential tariff treatment under T-MEC requires correct origin qualification and documentation. Many companies leave money on the table through incomplete origin processes or overly conservative classification. Periodic origin audits and classification reviews are high-ROI compliance activities.

IMMEX and free-trade-zone logic

IMMEX allows temporary import of inputs for export manufacturing with duty and tax benefits. Proper use of the program (and complementary programs) can significantly improve the cost position of export-oriented operations. Misuse creates serious compliance exposure.

Intermediate vs Advanced

Intermediate: executes cross-border shipments within established procedures and documentation requirements.**Advanced**: designs the trade-compliance architecture, maximizes legitimate preferential treatment, integrates compliance into procurement and network design, and manages the relationship with customs brokers and authorities.

BEST PRACTICES

Integrate trade compliance into product and supplier decisions

Origin and classification consequences should be visible at the time of sourcing and product design, not discovered at the border.

Treat IMMEX and similar programs as strategic assets

They require disciplined compliance but can materially improve the cost position of export manufacturing.

Audit classifications and origin claims periodically

Both under-claiming (leaving money on the table) and over-claiming (creating exposure) are common.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Nearshoring manufacturer moving volume from Asia to Mexico — Initial landed-cost models understated the impact of origin and IMMEX benefits.

Element	Asia Baseline	Mexico with full T-MEC + IMMEX optimization
Unit manufacturing cost	Lower	+8–12%
Duty + brokerage + inland	Higher and variable	Materially lower
In-transit inventory + safety stock	High (30–45 days)	Dramatically lower (1–4 days)
Total landed + inventory cost	Baseline	-9% to -15% depending on product

Result: Once origin qualification, IMMEX and inventory effects were correctly modeled, the nearshoring case became clearly positive for the majority of the portfolio. Several SKUs that looked marginal on pure manufacturing cost became attractive.

WHAT YOU WOULD USE

Trade & Customs

Trade management systems (Thomson Reuters ONESOURCE, E2open (formerly Amber Road), Descartes, etc.) — Classification, preference management and customs filing support.

Broker systems + internal compliance trackers — Day-to-day execution layer that must be tightly controlled and audited.

HOW IT IS MEASURED

T-MEC / Preferential Origin Utilization %

(Value of imports/exports correctly claiming preferential treatment / Eligible value) × 100

Benchmark: Target close to 100% of genuinely eligible flows

Customs Clearance Cycle Time

Average hours/days from arrival to release

Benchmark: Track by border crossing and mode; chronic outliers indicate process or broker issues

Treat customs as a design variable, not paperwork. Own classification and origin, model full landed cost, use programs like IMMEX deliberately and audit before the authority does.

05 Sustainable logistics: carbon footprint, green transport & circular supply chains

Sustainable logistics measures and reduces the emissions and waste of transport, warehousing and packaging — using recognized methods such as the GLEC Framework and ISO 14083 — and redesigns flows for lower carbon and, often, lower cost.

Sustainable logistics is increasingly a commercial and regulatory requirement, not only a values statement

Customers, investors and regulators are raising expectations on Scope 3 emissions, particularly those embedded in transportation and distribution. At the same time, many of the highest-impact decarbonization levers (route optimization, consolidation, modal shift to rail, higher fill rates) also reduce cost. The opportunity is to pursue the initiatives that deliver both.

Measurement is the foundation

You cannot manage what you do not measure. Calculating logistics emissions (well-to-wheel or tank-to-wheel according to chosen standard) by mode, lane and customer enables both internal prioritization and credible external reporting. Tools aligned with GHG Protocol and ISO 14083 (which superseded EN 16258) are becoming standard.

Modal shift and network decisions have the largest impact

Shifting appropriate volume from road to rail, improving consolidation, and reducing empty miles typically dwarf the impact of incremental vehicle-efficiency improvements. Network design and mode strategy are therefore the highest-leverage sustainability decisions in logistics.

Intermediate vs Advanced

Intermediate: tracks basic emissions metrics and supports reporting requests.**Advanced**: builds a credible logistics emissions baseline, identifies the highest-ROI reduction levers, integrates sustainability into network and mode decisions, and engages 3PLs and carriers on shared targets.

BEST PRACTICES

Measure before you set targets

A credible baseline is the prerequisite for any serious reduction program and for external claims.

Prioritize levers that improve both emissions and cost

Consolidation, fill-rate improvement, modal shift and network optimization usually sit in this quadrant.

Include emissions expectations in 3PL and carrier governance

What gets measured and discussed in QBRs gets managed. Sustainability that remains outside the commercial relationship remains aspirational.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Consumer goods company setting a logistics emissions reduction target — Baseline showed road freight dominating both cost and emissions.

Lever	Emissions Impact	Cost Impact
Improved consolidation + higher fill rates	-9%	-6% freight
Modal shift of suitable volume to rail	-14% on shifted lanes	Neutral to slightly favorable
Carrier scorecard including emissions intensity	Gradual improvement	Slight premium on some lanes
Packaging / unitization improvements	-3%	Favorable

Result: The company achieved a 17% reduction in logistics emissions intensity over three years while simultaneously reducing total freight cost. The initiatives that improved both dimensions were prioritized first.

WHAT YOU WOULD USE

Sustainable Logistics

EcoTransIT / NTMCalc / similar emission calculation engines — Widely used for transport emission calculations aligned with recognized standards.

Visibility platforms with emissions modules (e.g. project44) — Increasingly offer per-shipment or per-lane emissions tracking integrated with operational data.

HOW IT IS MEASURED

Logistics COe per Unit / per tkm

Emissions intensity of transportation and distribution activities

Benchmark: Track absolute and intensity; intensity is usually the more actionable management metric

% of Volume on Lower-Emission Modes

Share of freight moved by rail, intermodal or other lower-intensity modes

Benchmark: Depends on corridor suitability; the goal is to maximize appropriate shift

Measure with a standard, then cut what costs twice. Baseline by lane with GLEC / ISO 14083, prioritize levers that save both carbon and money, and make carriers part of the target.

Logistics Technology & Innovation

The technology layer transforming logistics from a labor-intensive operation into a data-driven, increasingly automated system — and the roadmap for capturing that value without over-investing in unproven technology.

01 Logistics technology stack: TMS, WMS & visibility platforms

The logistics technology stack is the set of execution systems — WMS for the warehouse, TMS for transport, visibility platforms for in-transit events — and the integrations that connect them to each other and to the ERP.

The logistics technology stack is only as strong as its integrations

Modern logistics runs on three core systems: WMS (warehouse execution and inventory truth), TMS (transportation planning, execution and settlement), and visibility platforms (multi-carrier, multi-mode tracking and exception management). Value is created when these systems exchange data cleanly with each other and with the ERP. Fragmented stacks produce re-keying, inconsistent inventory pictures and delayed decisions.

Native vs best-of-breed decisions

ERP-native modules (e.g., SAP EWM + SAP TM) offer superior integration and a single data model. Best-of-breed solutions often provide deeper optimization or industry-specific functionality. The correct choice depends on process complexity, existing ERP footprint and the organization's integration capability. Many companies end up with a hybrid and underestimate the ongoing integration cost.

Phased implementation is almost always superior

Attempting to replace WMS, TMS and visibility simultaneously creates extreme project risk. Recommended sequence in most environments: stabilize or implement WMS first (inventory foundation), then TMS, then the visibility/control-tower layer on top of clean execution data.

Intermediate vs Advanced

Intermediate: uses the available systems proficiently and surfaces functional gaps.**Advanced**: designs the target architecture, leads selection and integration, and ensures the stack actually changes process outcomes rather than merely digitizing the status quo.

BEST PRACTICES

Prioritize clean integration over feature richness

A simpler system that is fully integrated almost always outperforms a feature-rich system that requires constant manual intervention.

Sequence implementations deliberately

WMS -> TMS -> Visibility is the lowest-risk order in the majority of environments.

Budget seriously for change management and process redesign

Technology that is used as a recording system instead of a decision and optimization system delivers a fraction of its potential.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Distribution company with ERP + third-party WMS but no TMS or visibility

Gap	Impact	Solution
No TMS	Manual booking, unaudited invoices, low fill rates	SAP TM (native) or best-of-breed TMS
No multi-carrier visibility	Reactive exception handling, customer complaints	project44 / equivalent visibility layer
Manual ERP–WMS sync	Daily reconciliation effort, inventory discrepancies	Real-time integration

Result: Business case showed 9-month payback from headcount release, freight recovery and reduced expedites. Implementation sequenced WMS integration first, then TMS, then visibility.

WHAT YOU WOULD USE

Core Stack

SAP EWM + SAP TM — Strong choice when the company is already on S/4HANA and values native integration.

Manhattan / Blue Yonder / Körber (WMS) + specialized TMS + project44/FourKites — Common best-of-breed combination for complex operations.

HOW IT IS MEASURED

Systems Integration Rate %

(Logistics transactions processed without manual re-entry / Total logistics transactions) × 100

Benchmark: >85% is a realistic target for a well-integrated stack

Technology ROI (3-year)

(Quantified benefits – total cost of ownership) / total cost of ownership

Benchmark: Target >200–300% over three years for major stack investments

Buy integrations, not features. Assign data ownership, choose native or best-of-breed by complexity, sequence WMS -> TMS -> visibility, and fund the change management that makes systems change outcomes.

02 Warehouse automation: robotics, AGVs & goods-to-person systems

Warehouse automation replaces or assists manual handling with equipment and robotics — conveyors and sorters, AGVs and AMRs, goods-to-person systems, AS/RS — chosen according to the operation's order profile, volume and labor economics.

Automation is a business-case decision, not a technology decision

Goods-to-person, AGVs, autonomous mobile robots, sorters and AS/RS can dramatically improve productivity, accuracy and density. They also require significant capital, create new maintenance and talent requirements, and perform poorly when volume, mix or process stability is insufficient. The correct starting point is a rigorous business case, not a vendor demonstration.

Key decision variables

Labor cost and availability, order profile (lines per order, units per line), volume and seasonality, required accuracy, available building height/density, and payback horizon. High-SKU, low-lines-per-order e-commerce profiles favor goods-to-person. High-volume, repetitive unit-load movement favors AGVs or conveyors. Low volume or highly variable operations often remain better served by well-designed manual processes.

Total cost of ownership is frequently understated

Maintenance contracts, spare parts, software licenses, system integrators, and the cost of specialized technical talent routinely add 8–15% of capex per year. Omitting these items produces optimistic paybacks that later disappoint.

Intermediate vs Advanced

Intermediate: operates automated systems to defined standards and reports performance.**Advanced**: builds the business case, leads vendor selection and design, manages implementation, and continuously optimizes the automated operation.

BEST PRACTICES

Build the business case with full TCO

Include maintenance, software, integration and specialized labor. Optimistic cases that ignore these items create future credibility problems.

Match technology to order and volume profile

The best technology for a high-volume unit-load operation is rarely the best technology for a high-SKU, low-lines e-commerce operation.

Plan the talent transition explicitly

Automation reduces operational headcount but increases the need for technical and analytical skills. The organization must be redesigned, not just the building.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — E-commerce fulfillment center, 18,000 active SKUs, high labor turnover

Metric	Before (manual)	After (Goods-to-Person)
Picking productivity	68 lines/hour	240 lines/hour
Picking accuracy	98.4%	99.97%
Picking FTEs	180	45
Peak capacity constraint	Labor availability	Largely removed

Result: Capex $4.8M, payback ~2.8 years on labor and error reduction alone. The strategic benefit of reliable peak capacity was judged even more valuable than the pure financial return.

WHAT YOU WOULD USE

Warehouse Automation

AutoStore / Exotec / Swisslog / Knapp (GTP & AS/RS) — Leading goods-to-person and automated storage solutions.

Locus / Geek+ / other collaborative AMR vendors (AMR collaborative picking) — Lower-capex collaborative robot options that often deliver faster payback in mid-sized operations.

HOW IT IS MEASURED

Picking Productivity (lines or units per hour)

Output per labor hour or per robot hour

Benchmark: Manual 60–120 lines/h · GTP systems commonly 200–400+

Picking Accuracy %

(Correct lines / Total lines picked) × 100

Benchmark: >99.9% expected from mature automated systems

Fit the technology to the order profile and the business case. Stabilize the process, count the full TCO, test downside scenarios and redesign the organization along with the building.

03 Autonomous delivery & drone logistics

Autonomous delivery covers drones and autonomous ground vehicles that move parcels without a driver on board, currently viable in specific niches such as medical samples, remote access and closed campuses.

Autonomous delivery is real in specific niches; mass deployment is still constrained

Drone delivery and autonomous ground vehicles are past pure pilot stage in selected use cases (medical samples, remote/rural pharmacy, campus and closed-environment delivery). In open urban environments and for general e-commerce, regulatory, infrastructure, cost and public-acceptance barriers remain significant. In Mexico (2026) the regulatory framework (AFAC and related) is still evolving.

Where the economics already work

High-urgency, low-weight, point-to-point flows in areas with difficult ground access or severe congestion. Medical sample networks between hospitals, emergency pharmaceutical delivery to remote clinics, and certain industrial campus applications are the clearest current opportunities.

Realistic timeline for broader Mexican adoption

Scaled commercial drone or ADV networks for general last-mile in major Mexican cities are more likely a 2028–2032 phenomenon than a 2026–2027 one. Companies should monitor regulation and run focused pilots in high-value niches rather than assume imminent mass replacement of traditional last-mile.

Intermediate vs Advanced

Intermediate: stays informed on the state of the technology and its implications.**Advanced**: evaluates specific niche business cases, manages regulatory and partner relationships, and runs disciplined pilots with clear success criteria.

BEST PRACTICES

Focus pilots on niches where the value of time or access is extreme

General e-commerce last-mile is still a difficult economic and regulatory environment for autonomy.

Engage regulators early

In Mexico the AFAC framework is a critical path item. Technical feasibility without regulatory permission has no operational value.

Measure real TCO in the pilot, not brochure numbers

Pilot-phase costs are higher; the question is whether the mature-state economics and service benefits justify the journey.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Hospital network evaluating drones for urgent lab samples (CDMX area)

Criterion	Ground Courier	Drone (proposed pilot)
Average urgent delivery time	47 min (highly variable)	12–18 min (weather permitting)
Cost per delivery	$85 average	~$40 mature / ~$120 pilot phase
Reliability	Traffic dependent	Weather and airspace dependent
Regulatory readiness	Existing	Requires specific AFAC corridor approval

Result: Pilot approved for a limited high-urgency corridor after regulatory engagement. The business case is positive for the medical-sample use case even at pilot-phase costs because of the clinical value of time.

WHAT YOU WOULD USE

Autonomous Delivery

Wing (Alphabet) / Zipline / other specialized drone operators — Most mature commercial drone delivery platforms, especially in medical and remote applications.

Starship and similar ground ADV platforms — Relevant for campus, closed-community and certain urban sidewalk/road applications.

HOW IT IS MEASURED

Drone / ADV Delivery Success Rate %

(Successful autonomous deliveries / Attempted autonomous deliveries) × 100

Benchmark: >95% required for operational credibility; >99% expected in regulated medical use cases

Autonomous vs Traditional Cost per Delivery

Fully loaded cost comparison on the same lanes or use cases

Benchmark: Must reach clear advantage or strategic necessity to justify scale investment

Pilot where the value of time is extreme. Engage AFAC early, set success criteria upfront and compare honestly with the best ground alternative before scaling.

04 Digital twins & simulation in logistics network planning

A logistics digital twin is a data-fed model of the network — nodes, flows, capacities, inventories and rules — used to simulate changes and disruptions before they happen in the physical operation.

A digital twin lets you break the network without breaking the business

A logistics digital twin is a living digital representation of nodes, flows, inventories, capacities and constraints. It enables rapid what-if analysis of network changes, disruption scenarios and policy decisions before they are implemented in the physical world. Its value is proportional to the quality, granularity and freshness of the data that feeds it.

Highest-value use cases

Disruption simulation (DC outage, carrier capacity loss, border closure), network redesign evaluation, inventory policy stress-testing, and S&OP scenario comparison. The twin is particularly powerful when it can be updated with near-real-time operational data rather than static monthly snapshots.

Data quality is the real constraint

A sophisticated simulation engine fed with stale or inaccurate data produces confident but wrong answers. Investment in data pipelines and master-data quality often yields higher returns than investment in more advanced simulation algorithms.

Intermediate vs Advanced

Intermediate: uses existing simulation or scenario tools for specific analyses.**Advanced**: designs and maintains the digital twin, ensures continuous data feeds, and embeds twin-based scenario analysis into regular S&OP and network governance processes.

BEST PRACTICES

Invest in data pipelines before exotic simulation features

Garbage-in-garbage-out is especially dangerous when the output looks precise and authoritative.

Start with the highest-pain scenarios

DC outage, key carrier failure and border disruption usually deliver the fastest insight and organizational learning.

Keep the twin synchronized with reality

A twin that still contains a closed facility or an obsolete lane produces misleading recommendations.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Pharmaceutical distributor stress-testing a 72-hour main-DC outage

Scenario Element	Simulated Impact	Prepared Response
Main DC offline 72 h	32% of volume impacted on days 1–2	Pre-defined allocation to secondary nodes
Secondary DC capacity	Covers ~65% of critical demand	Priority rules by clinical criticality
Additional carrier capacity	Confirmed in simulation	Pre-negotiated surge agreements activated
Customer communication	—	Proactive messaging templates ready

Result: When a real (shorter) disruption later occurred, the team executed the pre-tested plan. Customer impact was estimated at less than 25% of the no-plan scenario.

WHAT YOU WOULD USE

Digital Twin & Simulation

Kinaxis Maestro (formerly RapidResponse) / Blue Yonder Luminate — Planning platforms with strong scenario and digital-twin capabilities.

anyLogistix / specialized simulation tools — Used for detailed network and operational simulation studies.

HOW IT IS MEASURED

Digital Twin Data Latency

Maximum age of the operational data reflected in the twin

Benchmark: <4 h for operational twins · <24 h acceptable for tactical planning twins

% of Major Network Decisions Supported by Simulation

(Significant decisions that used twin/scenario analysis / Total significant network decisions) × 100

Benchmark: >80% indicates the twin has become part of the management system

Build the twin around real questions and fresh data. Start with a high-pain scenario, automate the data, validate on history and use it where decisions are actually made.

05 AI & machine learning applications in logistics

AI and machine learning in logistics apply prediction and optimization to operational data — late-shipment prediction, dynamic ETAs, freight invoice anomaly detection, capacity forecasting — to support the people who make logistics decisions.

AI in logistics creates value in prediction, optimization and exception detection

Practical, high-ROI applications today include: carrier and lane OTIF prediction, dynamic ETA improvement, demand-aware capacity forecasting, anomaly detection in freight invoices, and reinforcement-learning-assisted routing. The limiting factor is almost never the algorithm; it is the quality, volume and accessibility of historical operational data.

Start where feedback loops are short and value is measurable

Predicting which shipments are likely to be late and intervening before the customer notices produces visible service and cost benefits quickly. Large, ambitious “AI transformation” programs that begin with poorly instrumented processes tend to stall.

Limitations that must be respected

Models trained on historical data struggle with unprecedented disruptions. Bias in historical decisions can be amplified. Adoption fails when planners do not trust or understand the recommendations. AI should augment human decision-makers, not attempt to replace contextual judgment.

Intermediate vs Advanced

Intermediate: uses AI-generated insights and predictions provided by platforms.**Advanced**: identifies high-value use cases, ensures data readiness, leads pilot design and evaluation, and drives adoption so that model output actually changes decisions.

BEST PRACTICES

Begin with use cases that have short feedback loops and clear ROI

Late-shipment prediction, invoice anomaly detection and dynamic ETA improvement are reliable starting points.

Treat data quality as the primary investment

No algorithm overcomes systematically incomplete or incorrect historical data.

Design for human + AI collaboration

Planners who understand why a recommendation was made are far more likely to act on it and to provide the feedback that improves the model.

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Distribution company implementing carrier OTIF prediction

Application	Model Performance	Operational Impact
OTIF prediction by carrier + lane + day	AUC 0.84 · 78% accuracy on delay prediction 48 h ahead	Enabled proactive reassignment
Proactive reassignment of critical shipments	—	+4.2 pp OTIF on high-risk days
Automatic alerts to customer service	—	68% of delays communicated before customer contact

Result: Customer complaints related to late delivery fell 28% in six months. The majority of the gain came from acting on predictions rather than from the predictions alone.

WHAT YOU WOULD USE

AI in Logistics

project44 AI Insights / FourKites Intelligent Visibility — Strong predictive ETA and risk capabilities built on large visibility data sets.

Blue Yonder Luminate / Kinaxis / custom ML on cloud platforms — Broader planning and optimization AI capabilities for companies ready to invest in data foundations.

HOW IT IS MEASURED

AI Decision Support Rate %

(Decisions of a given type that used AI input / Total decisions of that type) × 100

Benchmark: >60% indicates meaningful adoption for the targeted decision type

Value Realized from AI Use Cases

Quantified service or cost improvement attributable to AI-supported decisions

Benchmark: Track per use case; avoid vanity metrics such as “number of models deployed”

Start narrow and measure decisions, not models. Pick use cases with short feedback loops, fix the data first, pilot with a control group and design for planner trust.

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